

W01

AI Foundations and Python Orientation

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ADD 5 TO 7 LEARNING OUTCOMES

By the end of this session, students should be able to:

- Explain artificial intelligence in simple terms.
- Distinguish AI, machine learning, deep learning and generative AI.
- Identify common AI domains and real-world applications.
- Describe the main stages of an AI development lifecycle.
- Discuss common challenges, including data, bias, privacy and reliability.
- Explain why Python is widely used for learning and developing AI systems.

Key goal

Build a clear foundation before studying Python programming and AI models in detail.

Week 01 Learning Map

W01

- 01** AI Foundations

- 02** Development of AI

- 03** AI System Ecosystem

- 04** Real-World AI Challenges

- 05** Current Directions in AI

- 06** Python Orientation

SOURCE / FURTHER READING

01

AI Foundations

Huawei HCCDA-AI

What Looks Like Artificial Intelligence?

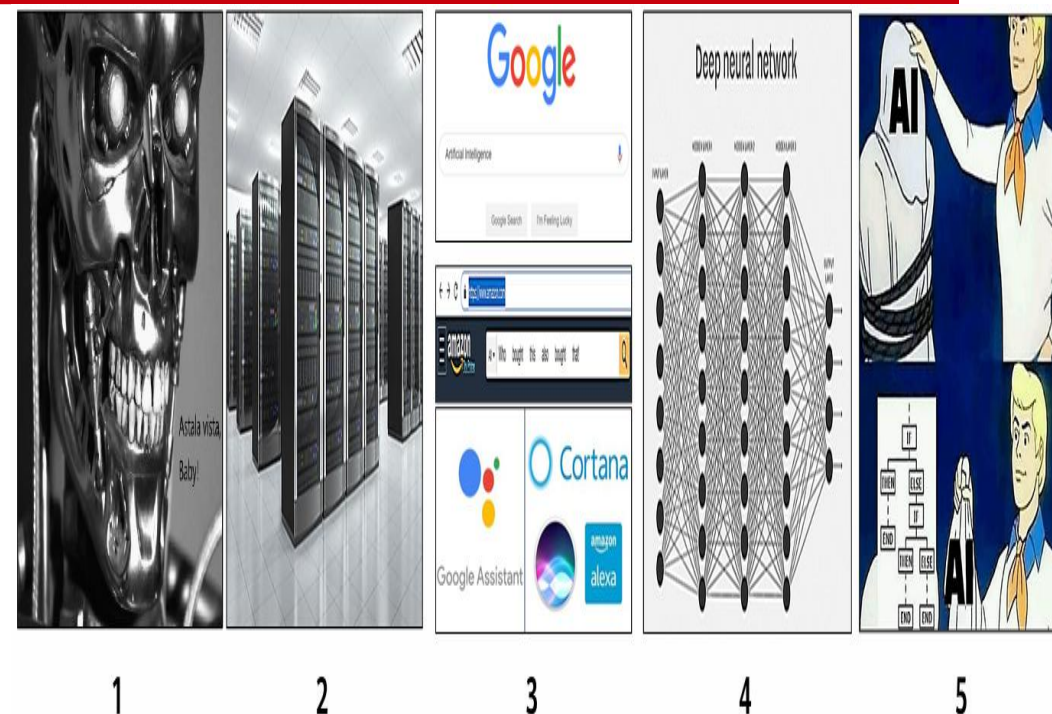
W01

These images show different ideas commonly associated with AI:

- **Terminator:** A fictional representation of intelligent machines.
- **Data centre:** Computing infrastructure that can run AI systems.
- **Intelligent applications:** Software that may use AI to perform useful tasks.
- **Neural network:** One method used in machine learning.
- **If-else rules:** Instructions written directly by a programmer.

Key point

AI is not one machine, application or algorithm. It is a broad field containing different methods, systems and technologies.



1

2

3

4

5

Terminator

Hyperscale
data center

Intelligent
apps

Neural
network

if-else

What Is Artificial Intelligence?

W01

Artificial intelligence is the field of building computer systems that perform tasks involving:

- Perception of images, sounds or sensor data
- Understanding and producing language
- Recognising patterns and making predictions
- Reasoning or selecting an appropriate action
- Generating text, images, audio or other content

AI systems may learn patterns from data, follow programmed rules, or combine both approaches.

Key point

An AI system can perform an intelligent-looking task without understanding the world in the same way as a human.



SOURCE / FURTHER READING

Input

Images, text, audio, measurements or sensor readings

Processing

Programmed rules or a model that has learned patterns from data

Output

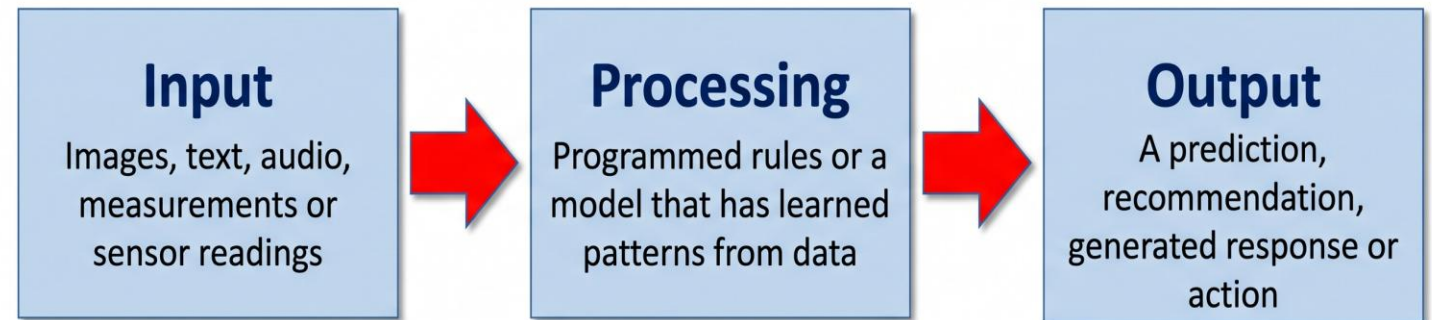
A prediction, recommendation, generated response or action

Example

A plant image enters the system. A trained model examines its visual patterns and predicts whether the plant is healthy or diseased.

Key point

The quality of the output depends on the data, processing method and evaluation of the system.



Automation

Automation follows instructions defined in advance.

Example

A timer switches a light on every day at 6:00 PM.

Artificial Intelligence

A learning-based system estimates patterns from examples.

Example

A model uses previous sensor readings to predict when lighting is needed.
Its prediction can change when the input data changes.

Key difference

Automation asks:

“Which instruction should be followed?”

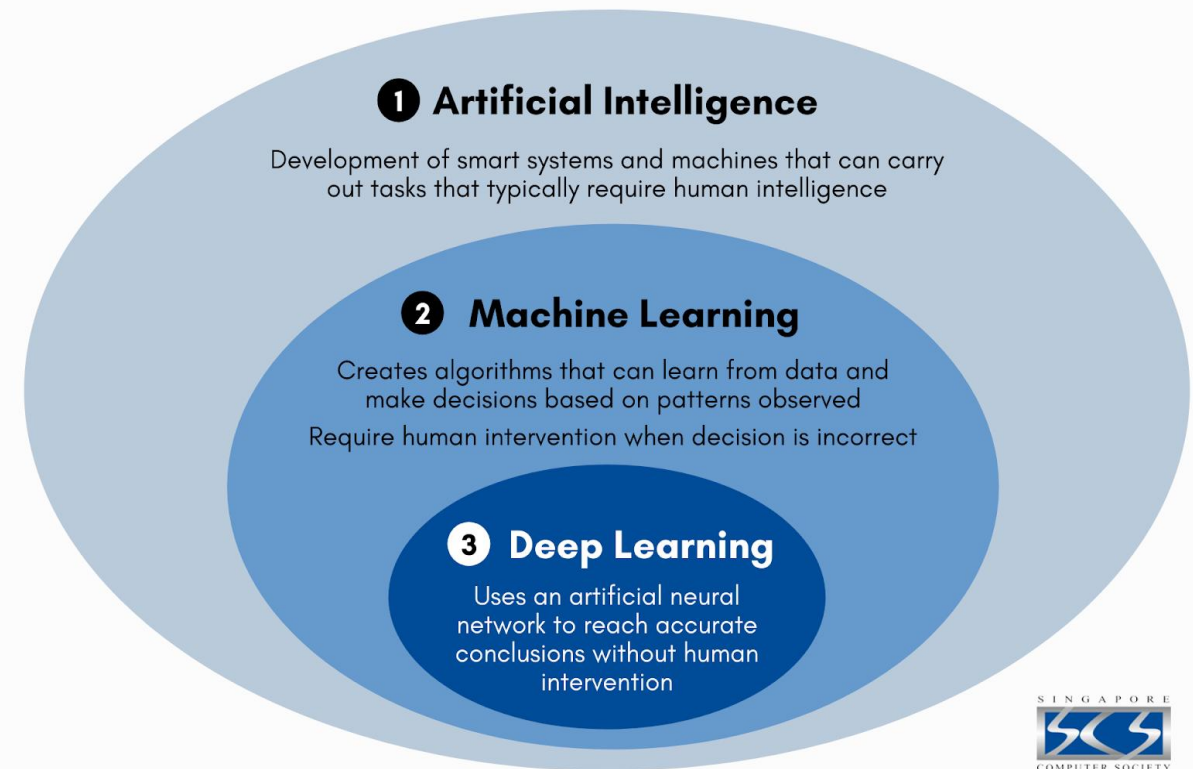
Learning-based AI asks:

“What pattern can be estimated from the available data?”

Some real systems combine programmed rules and learned models.

- **Artificial intelligence:** The broad field of creating systems that perform intelligent tasks.
- **Machine learning:** A part of AI that learns patterns from data.
- **Deep learning:** A part of machine learning that uses neural networks with multiple layers.
- **Generative AI:** Systems that produce new text, images, audio, video or code from learned patterns.

ARTIFICIAL INTELLIGENCE VS MACHINE LEARNING VS DEEP LEARNING



SOURCE / FURTHER READING

Narrow AI and AGI

W01

Narrow AI

Narrow AI is developed to perform a defined task or a limited group of related tasks.

Examples

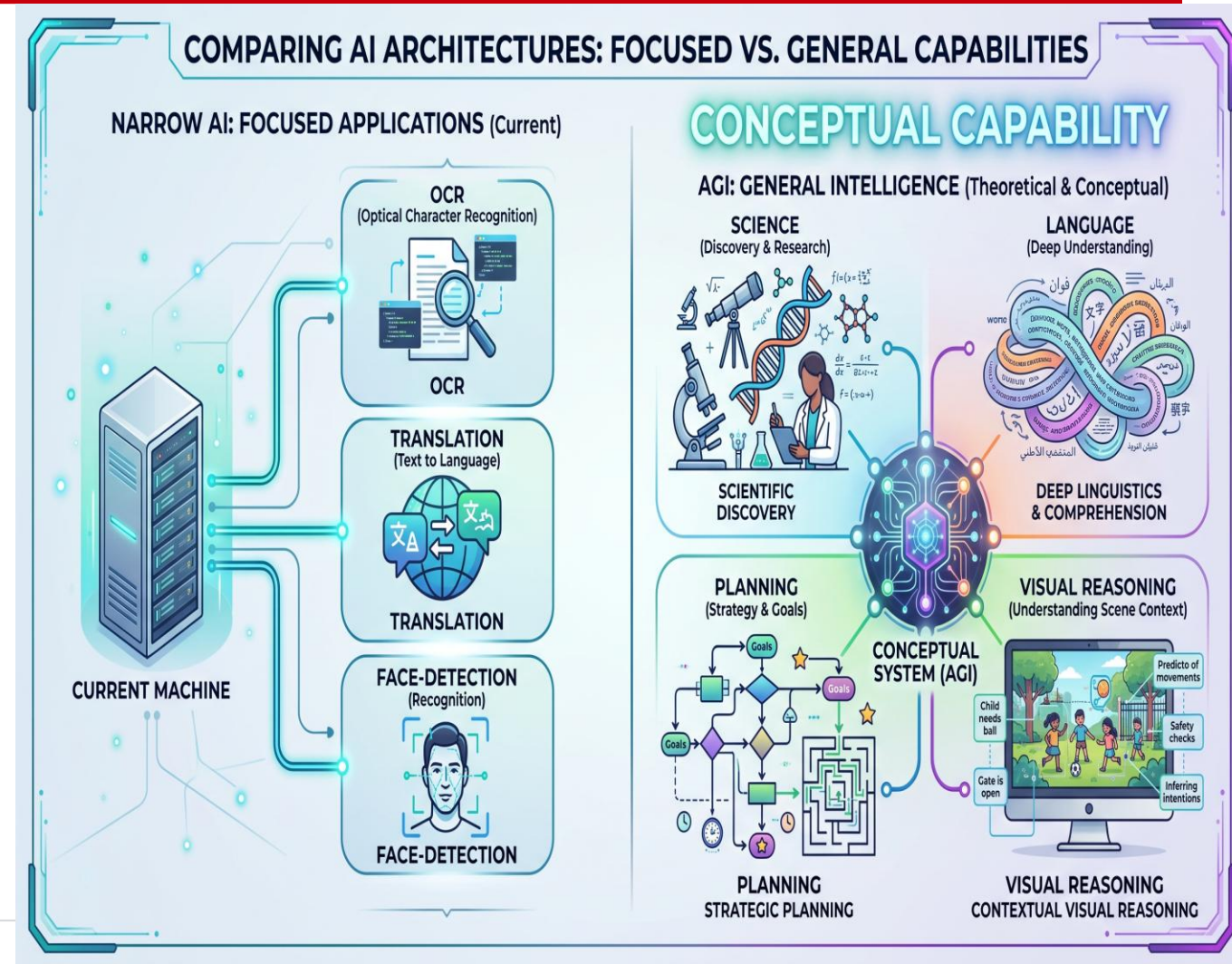
- Reading text from an image
- Detecting objects
- Translating languages
- Recommending videos
- Recognizing speech

AGI

- **What:** A proposed AI with human-like, broad intelligence.
- **Ability:** Learns many different tasks and transfers knowledge between them.

Key difference

Narrow AI performs defined tasks. AGI describes a much broader proposed capability.



Symbolic

Symbolic AI represents knowledge through explicit rules, logic and symbols.

Example

A laboratory safety system uses the rule:

If the temperature exceeds the safe limit, activate the warning.

The developer writes the rule directly.

Learning-Based AI

Machine learning estimates patterns from examples instead of receiving every rule directly.

Example

A model examines previous sensor measurements and learns to predict unusual laboratory conditions.

Its behaviour depends on the data used during training.

Key difference

Symbolic AI uses explicit knowledge and rules. Machine learning estimates patterns from data.

Modern systems may combine both approaches.

Major AI Domains

W01

- **Computer Vision**

Understands images and video

- **Natural-Language Processing**

Processes written language

- **Speech AI**

Recognizes or generates spoken language

- **Recommendation Systems**

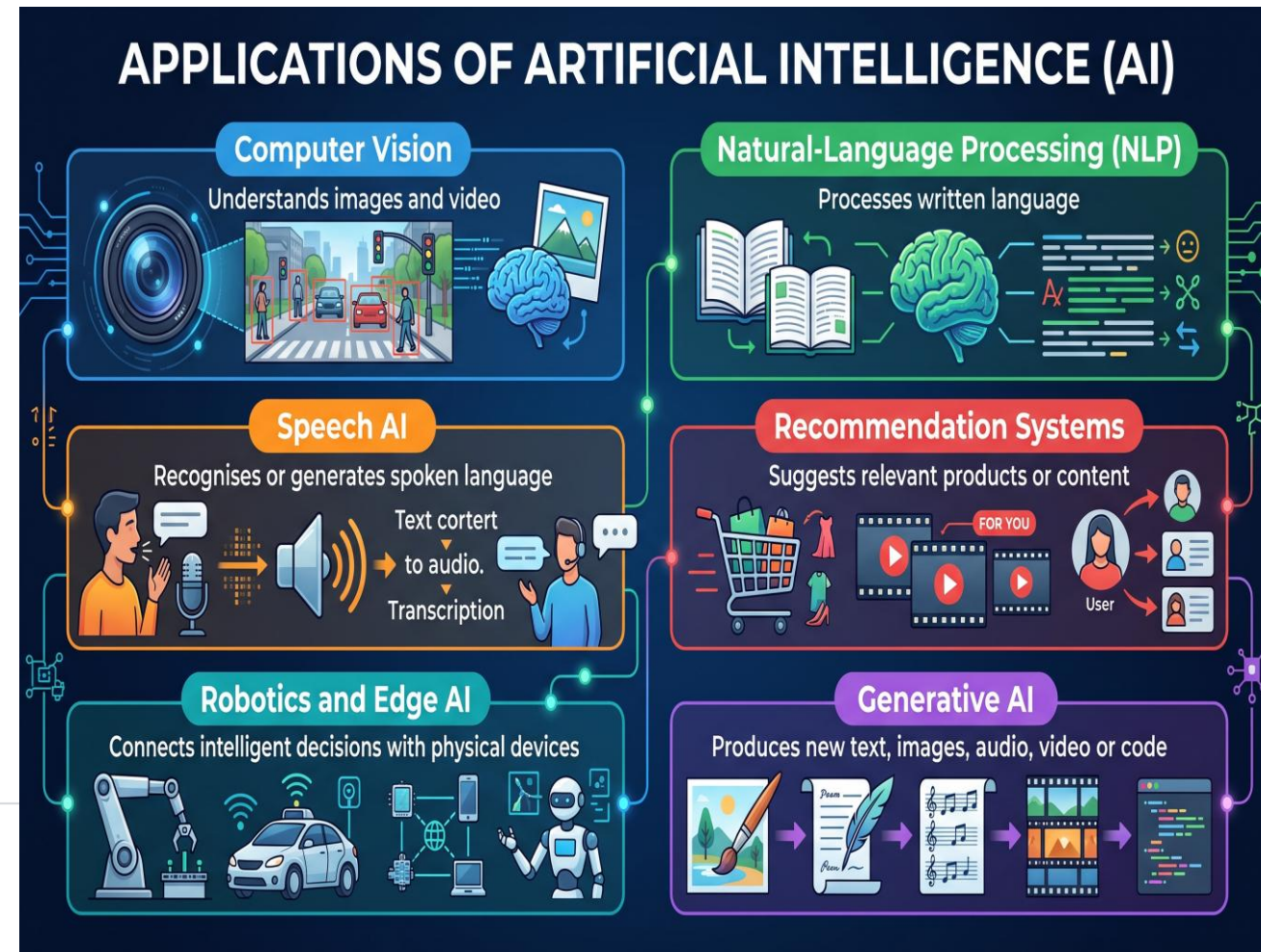
Suggests relevant products or content

- **Robotics and Edge AI**

Connects intelligent decisions with physical devices

- **Generative AI**

Produces new text, images, audio, video or code



02

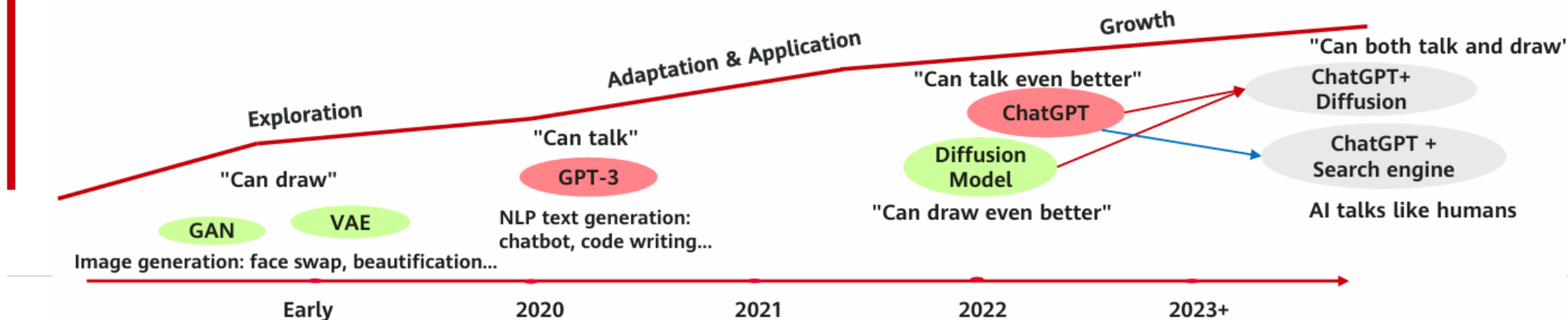
Development of AI

Huawei HCCDA-AI

AIGC: Artificial Intelligence Generated Content

Image generation: Typical image generation models are based on Diffusion models.

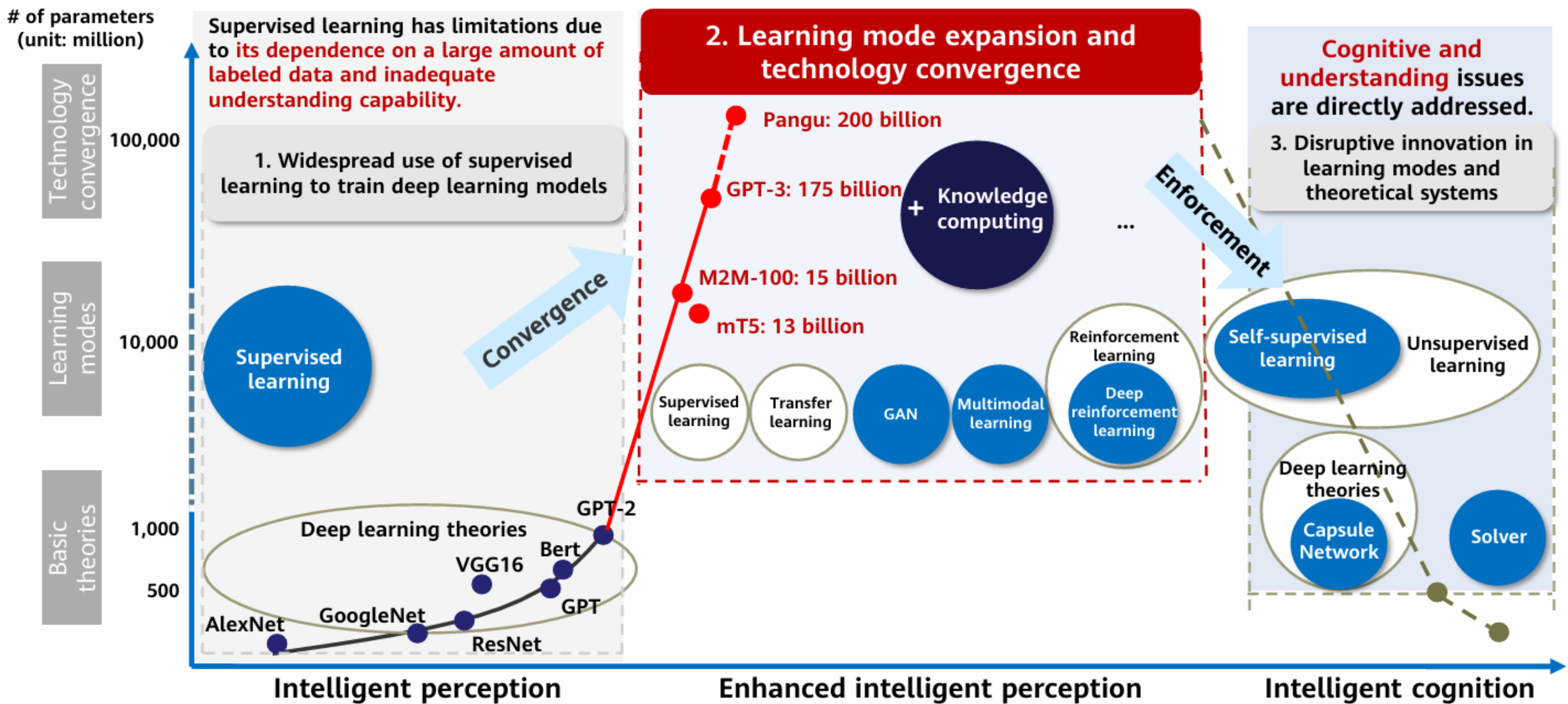
Natural language dialog: ChatGPT is a typical representative model, which is Transformer-based.



According to QbitAI's forecast, **text-to-image/video AI models** are expected to have the largest market share by **2030**. **China's AIGC market size is expected to exceed CNY1 trillion by 2030.**

AI Technology: From Intelligent Perception to Intelligent Cognition

W01



Today Analytical AI: Learning from data

- **Characteristics:** Analytical AI utilizes machine learning algorithms to **learn the underlying data distribution** and to classify, predict, group things and detect exceptions.
- **Typical technologies:** classical machine learning, fine-tuning, and neural networks such as convolutional neural network (CNN), deep residual network (DRN), and recurrent neural network (RNN)
- **Typical applications:**
 - ✓ Recommendation system: mining associations between users and items
 - ✓ Facial recognition: identifying a person based on facial feature information
 - ✓ Text recognition: recognizing text based on images containing text
 - ✓ Autonomous driving: traffic status monitoring and automatic parking



Tomorrow Generative AI: Learning from data + Generating data

- **Characteristics:** Generative AI **generates new data samples** by **learning the underlying data distribution and understanding how data is generated**.
- **Typical technologies:** Transformer, Diffusion, and Prompting
- **Typical applications:**
 - ✓ Text creation: generating a complete copy based on prompt text
 - ✓ Code generation: generating complete code based on the context
 - ✓ Image generation: generating images in various styles based on key information, such as those used in blogs and posters
 - ✓ Multi-modal generation: text-to-image generation



AIGC: AI Generated Content

03

AI System Ecosystem

Huawei HCCDA-AI

AI Technology Architecture

W01

Industries

Finance/Healthcare/Industry/Agriculture/Security
/Communication/Transportation/Game/VR & AR/UAV/Other

Applications

Image
recognition

Image
processing

Image
understanding

Audio
recognition

Text to
speech

Machine
translation

Semantic
understanding

Sentiment
analysis

Technologies

Computer
vision

Speech
processing

NLP

Planning and
decision system

Big data
analytics

Machine/Deep
learning
framework

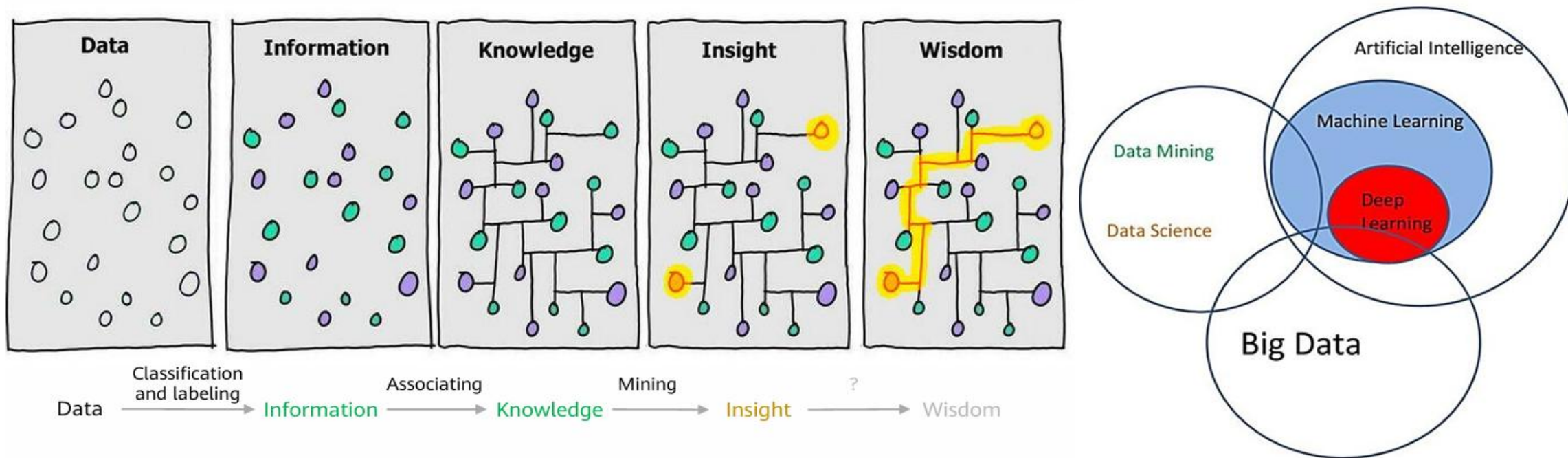
MindSpore, TensorFlow, PyTorch, and Caffe
(Alleviating the burden of programming and algorithm implementation)

Algorithms
(Model
training)

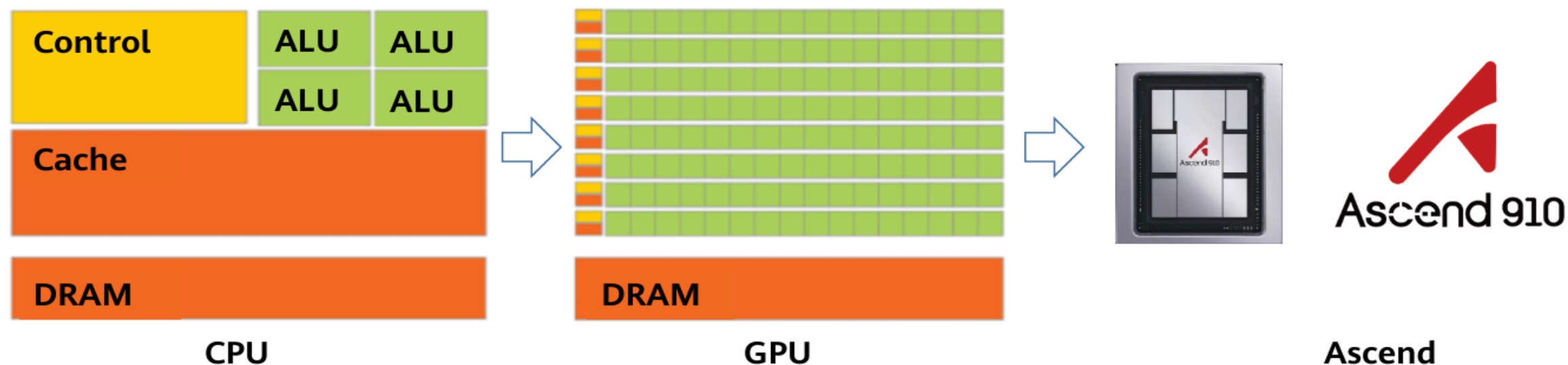
Machine learning, deep learning, and other algorithms; model training (big data + algorithms + optimization)

Infrastructure

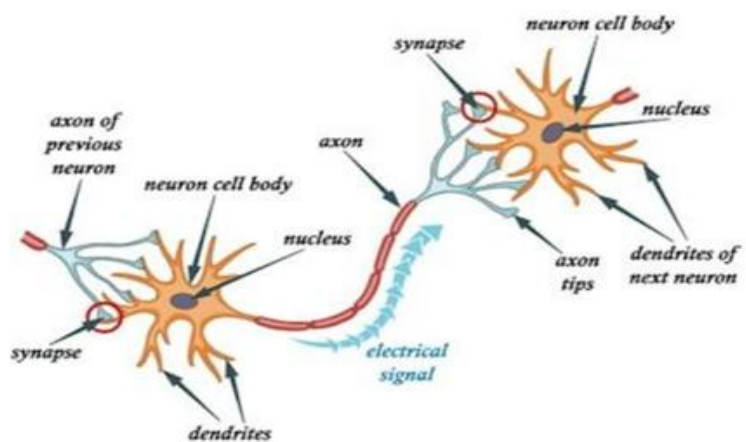
Computer hardware (Ascend chip and GPU), computing power, device deployment, and big data



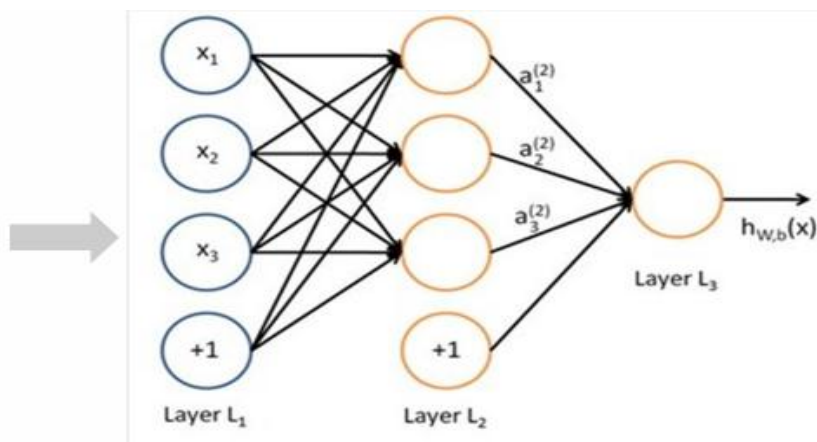
Data is an abstract expression of different phenomena in the real world. An enormous amount of data (big data) is generated every second worldwide from various sources such as netizens, mobile device users, machines, servers, labs, sensors, and devices. We can process this vast amount of big data and mine valuable information, knowledge, and even wisdom from it. Various AI services can utilize the acquired knowledge.



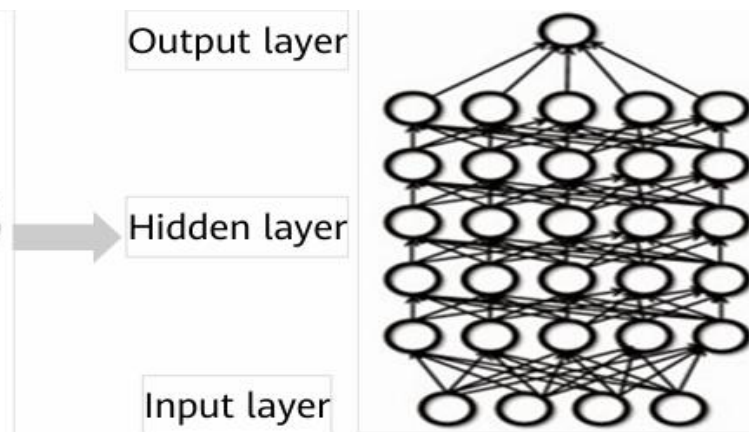
We've seen a shift from CPUs and GPUs to dedicated AI chips, such as Ascend AI chips, to provide significantly increased computing power. While GPUs are more versatile, dedicated AI chips have a simpler architecture and more advanced capabilities.



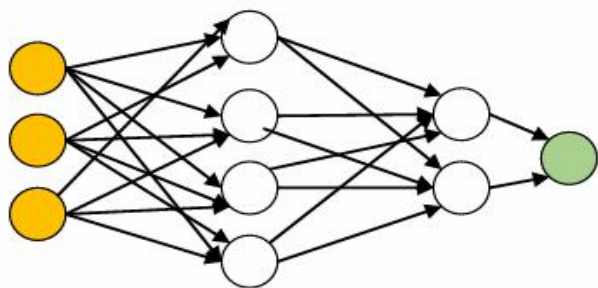
Human neural network



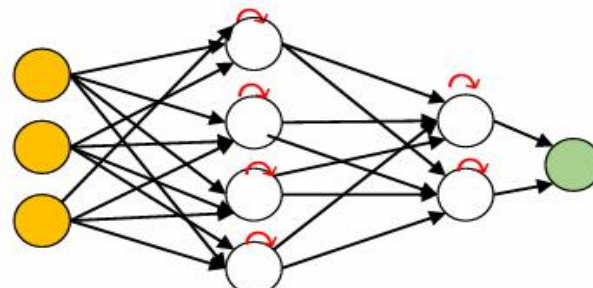
Perceptron



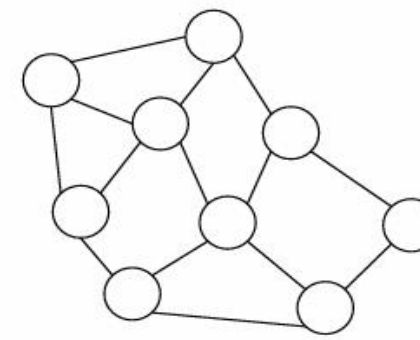
Deep neural network



(a) Feedforward neural network (FNN)



(b) Feedback network



(c) Graph network

- AI development frameworks are becoming easier to use and more versatile, continuously simplifying AI development. Examples of such frameworks include MindSpore, TensorFlow 2.x, and PyTorch.





Content moderation

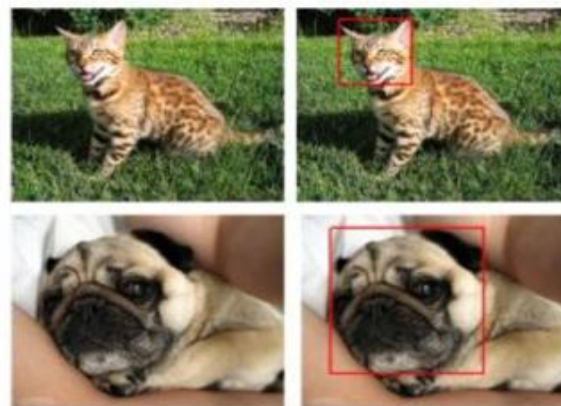
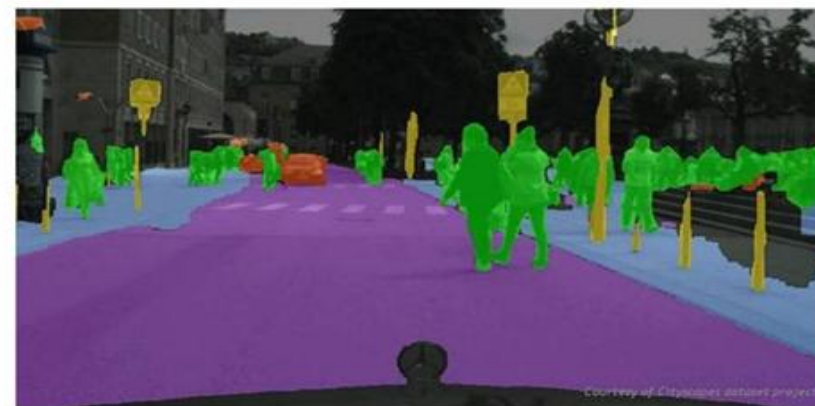


Image search



Semantic segmentation



Face detection



Object detection



Image classification

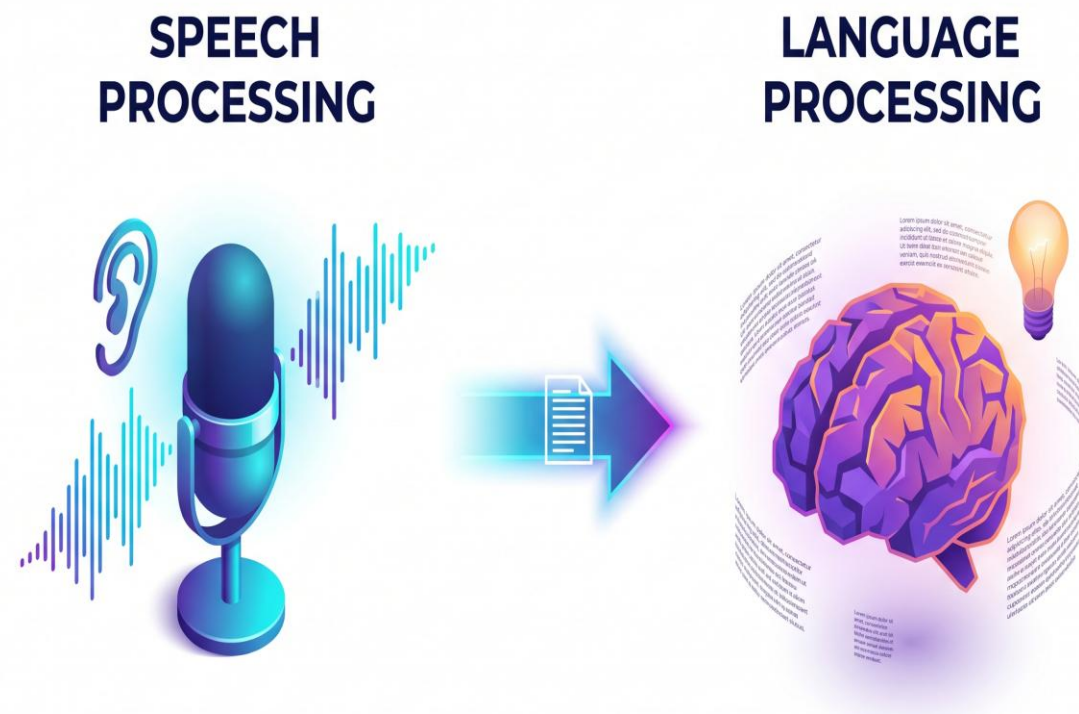
Speech Processing = The Ears (Sound)

- Focus:** Audio waves.
- Job:** Converts voice to text, or text to voice.
- Example:** Siri hearing your voice and typing it out.

Language Processing = The Brain (Meaning)

- Focus:** Text and context.
- Job:** Understands what the text actually means and decides how to respond.
- Example:** Siri reading the typed text, understanding you want the weather, and finding the answer.

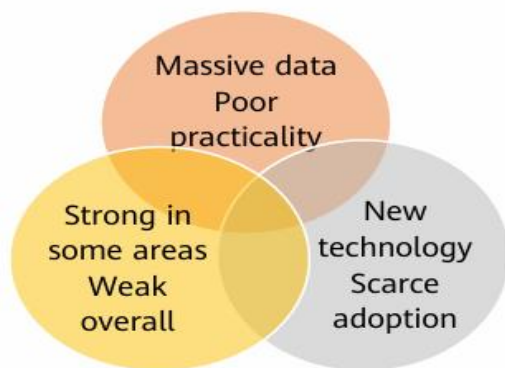
In one sentence: Speech processing hears the words; Language processing understands them.



Applications — AI in Smart Cities



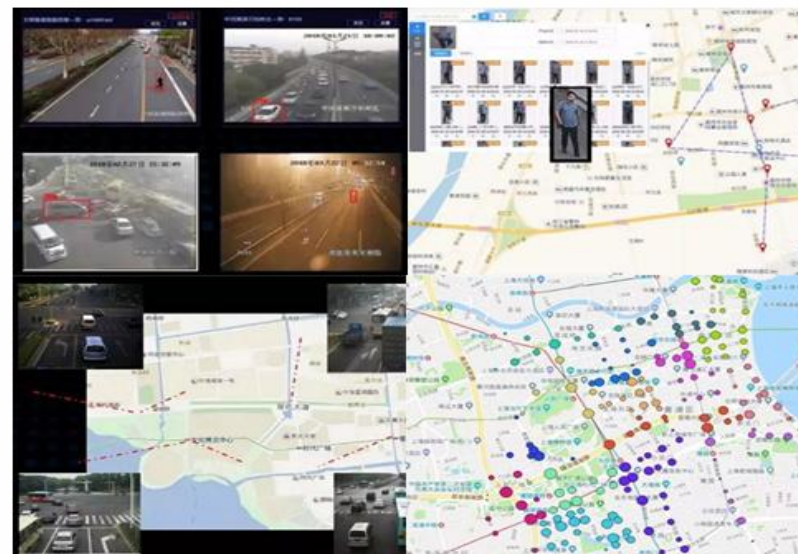
Excessive hardware but insufficient data



City-wide perception

City-wide warning

City-wide traffic dashboard



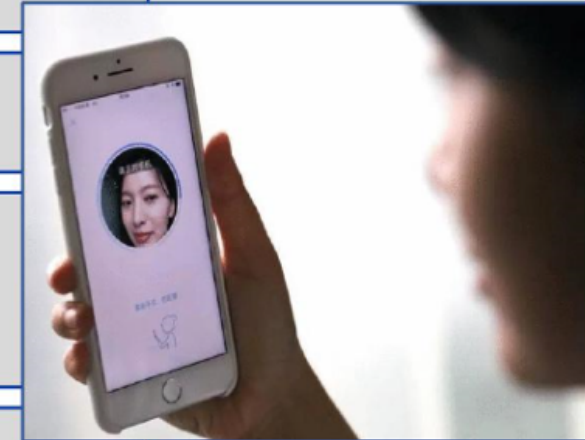
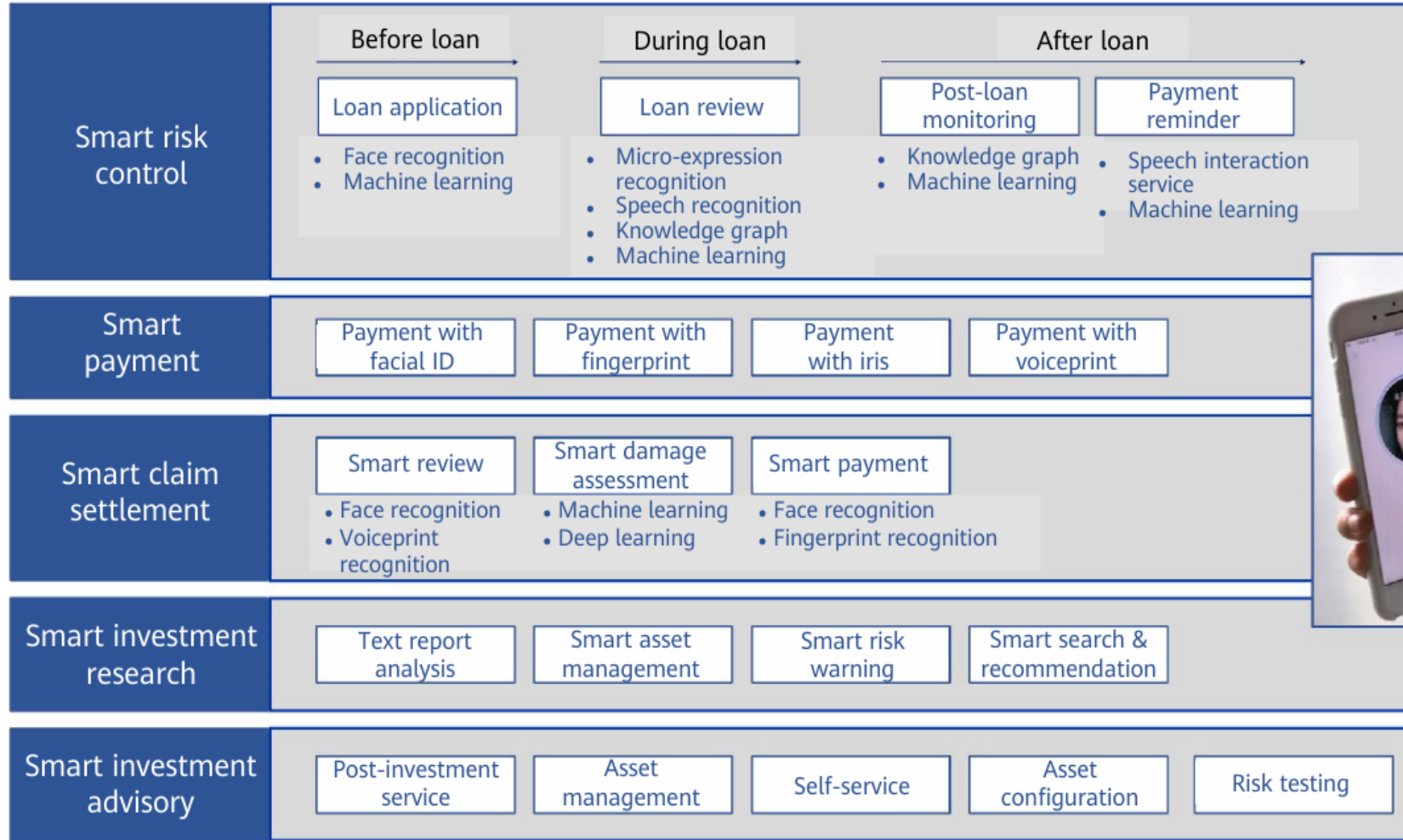
City-wide search

City-wide collaboration

Emergency vehicle monitoring



Applications — AI in Finance



Applications — AI in Medicare



Case recording via voice

- Efficiently records doctor-patient communication.
- Drives the digitalization of medicare.

Medical imaging analysis

- Automatically identifies and marks lesions.
- Helps doctors easily detect rare symptoms.

Comprehensive treatment

- Serves as a virtual doctor.
- Uses NLP, knowledge graphs, and CV to facilitate diagnosis and treatment.

Physical health management

- Monitors and manages health status, and predicts the occurrence of diseases.

Medical robots

- Examples: surgical robots and rehabilitation robots.
- Improves surgical precision.

Drug R&D

- Accelerates the discovery of candidate drugs.
- The technology, once matures, is expected to cut the cost of new drug R&D by US\$28 billion annually.

Applications — AI in Agriculture



Intelligent agriculture data analysis and predictions



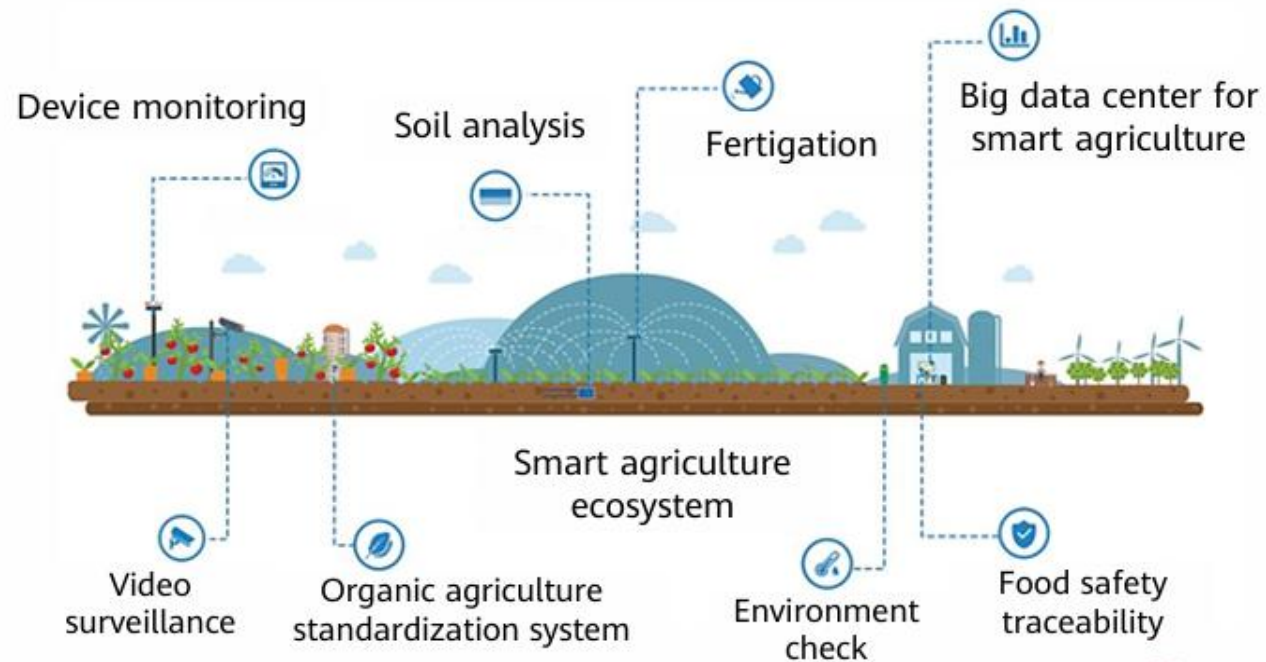
Pesticide cost reduction



Harvesting cost reduction



Airborne control system



04

Real-World AI Challenges

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Data Acquisition (Getting Data)

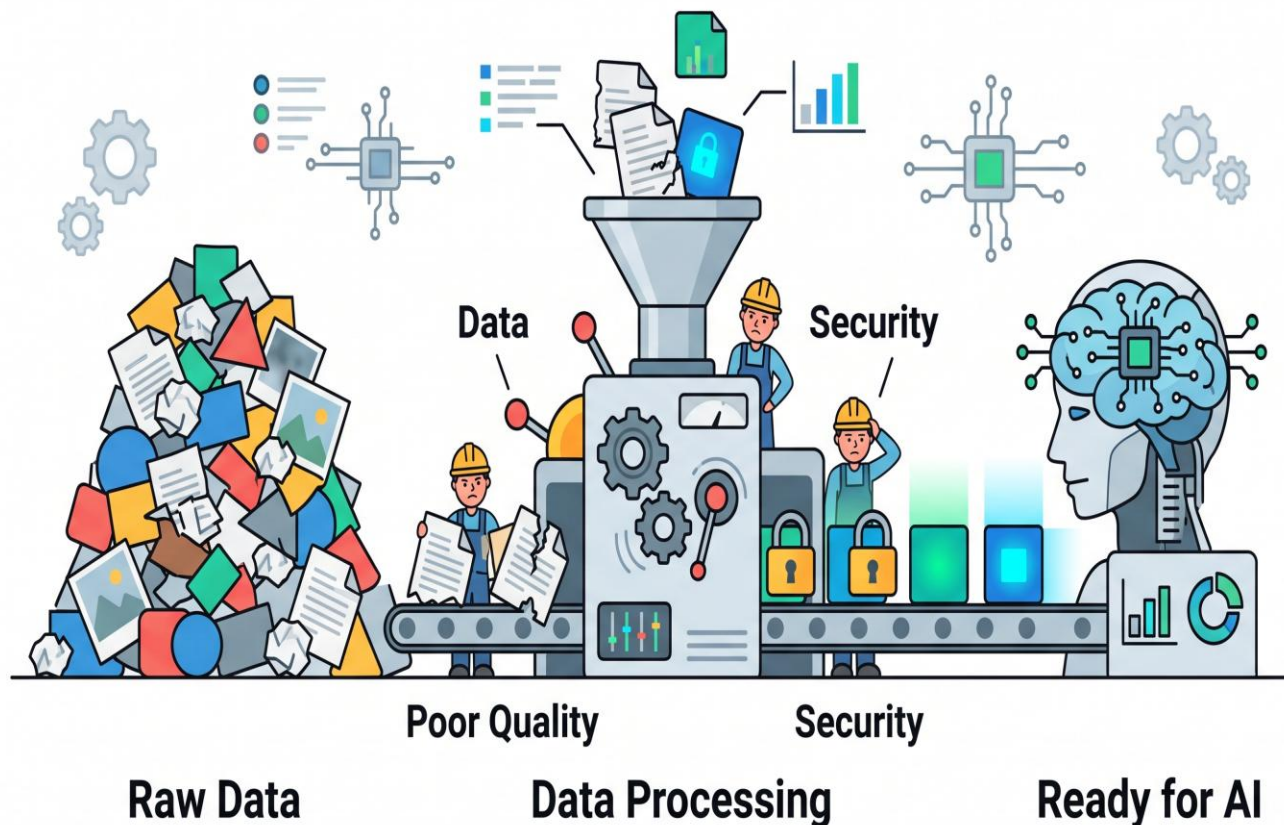
- Collecting raw information.
- Sources: Web, apps, sensors.
- Challenge: Scattered and expensive.
- Analogy: Gathering ingredients.*

Data Governance (Rules)

- Setting rules for data use.
- Focus: Privacy, laws, security.
- Challenge: Protecting data while using it.
- Analogy: The rulebook.*

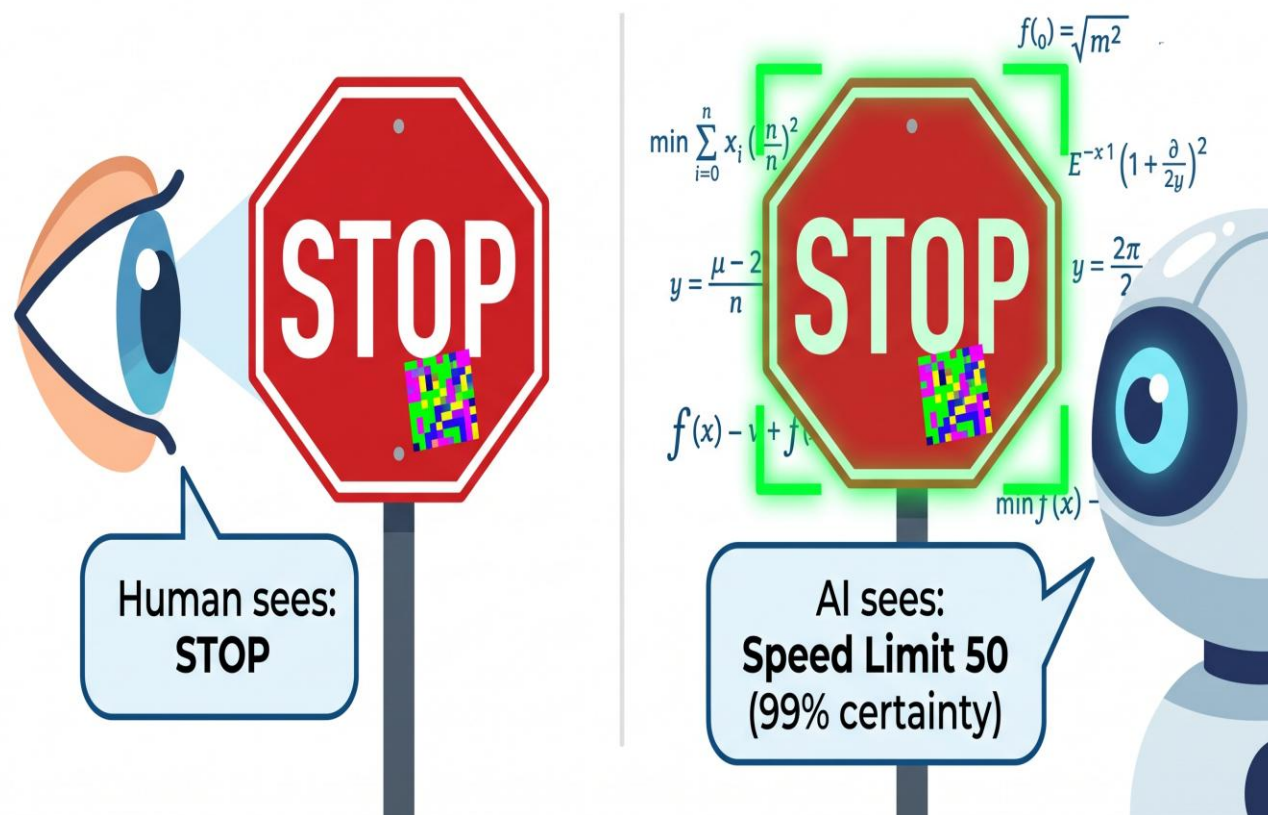
Data Labeling (Tagging)

- Tagging data so AI understands.
- Example: Drawing boxes around cats.
- Challenge: Slow, costly, hard for rare cases.
- Analogy: Flashcards for a toddler.*



- **Main Concept:** Adversarial Attack = Tricking AI with visuals, not code.
- **Human vs. AI:** Humans see context; AI sees only pixel math.
- **Magic Sticker:** A calculated pattern designed to break the AI's math.
- **Result:** AI confidently misreads STOP as "Speed Limit 50".
- **Danger:** Self-driving cars would ignore the stop sign and crash.
- **Takeaway:** AI lacks true understanding; it just matches fragile patterns.

Student for Mind, AI

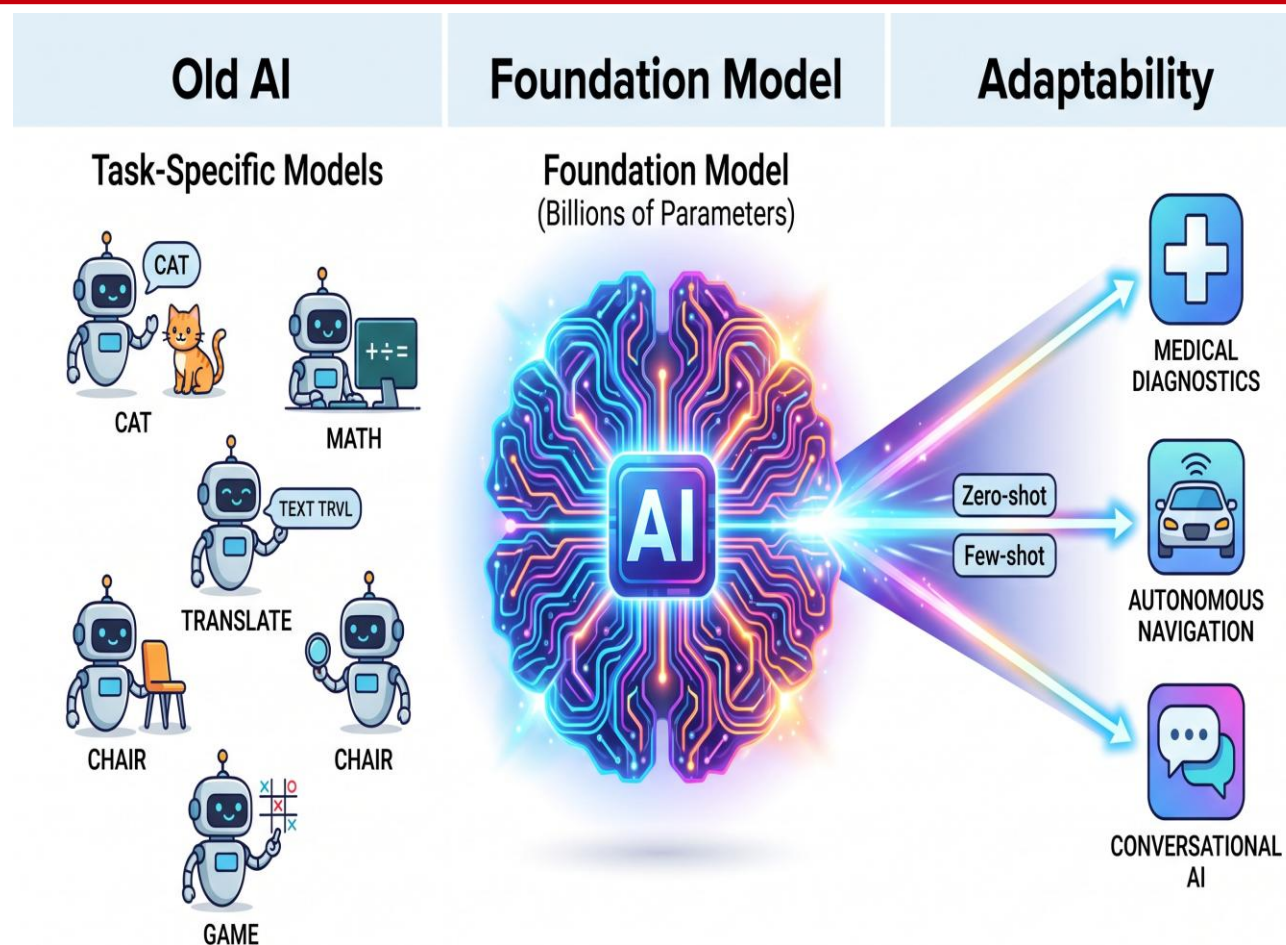


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Current Directions in AI

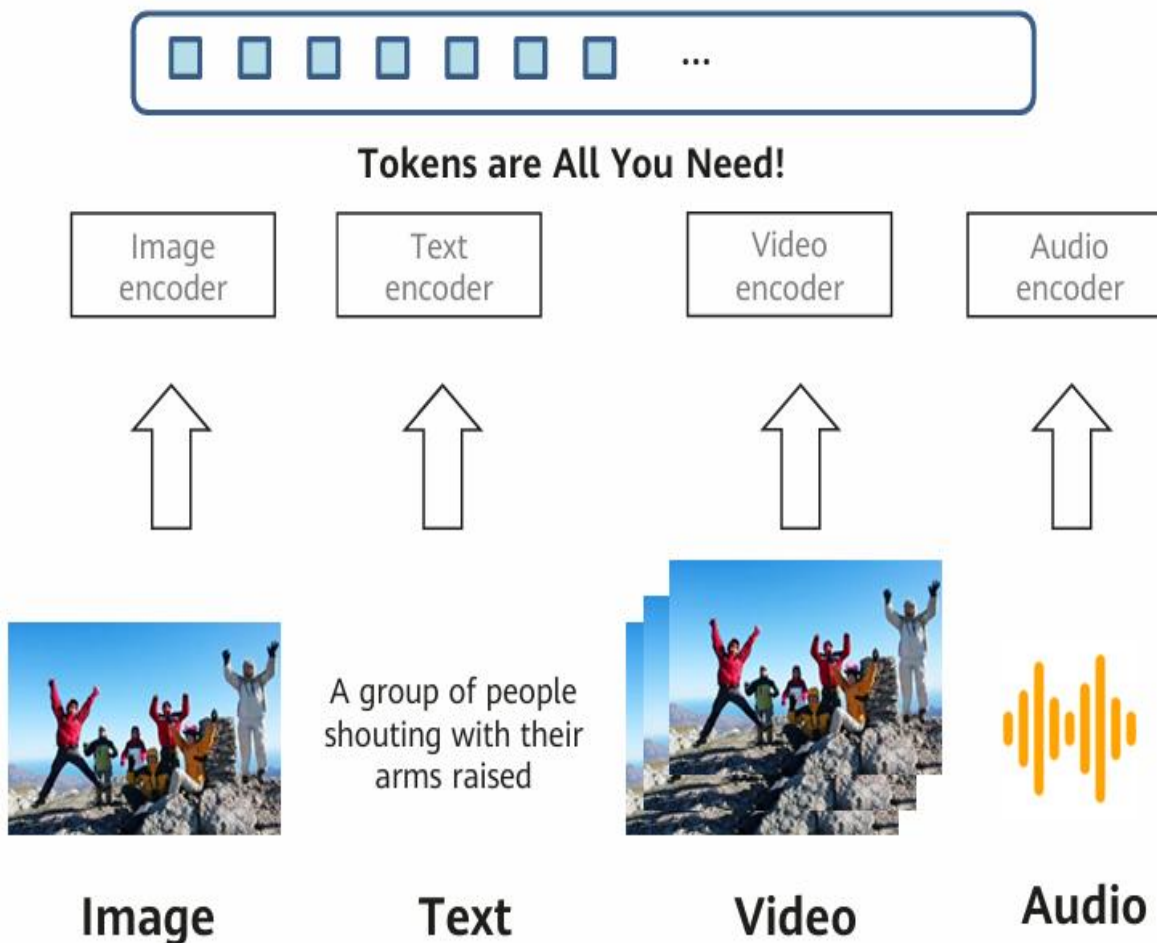
Huawei HCCDA-AI

- **What is it?** A "Big Brain" AI model with billions (even trillions) of internal connections (parameters). Most today are Large Language Models (LLMs) like ChatGPT.
- **How was it built?** Pre-trained on a massive, diverse pile of data (the whole internet). It learned general knowledge first.
- **Why is it special?** Old AI could only do *one* task (like translating). A Foundation Model is a "Jack of all trades" that can learn new tasks easily.
- **The Magic Trick (Zero/Few-shot):** You don't need to retrain it for new things. You can ask it to do something it's never done before (zero-shot), or just give it 2-3 examples (few-shot), and it gets it right.
- **The Benefit:** It saves time and money. Developers just plug it in to build apps for understanding, decision-making, and content generation.



Unified Multimodal AI

- **Language is Key:** Natural language is how humans describe the world.
- **The Problem:** AI handles different data types (text, images, audio) separately.
- **The Bridge:** Use language as the "central connector" or universal translator.
- **Unified Space:** Map all data types into one "mental map" of meaning.
- **The Result:** One single AI model can understand and do many different tasks.



☁️ Cloud AI (Huawei Cloud)

- What:** Massive server farms running the biggest AI models.
- Pros:** Unlimited power, smartest models (like Pangu).
- Cons:** Needs internet, slight delay.

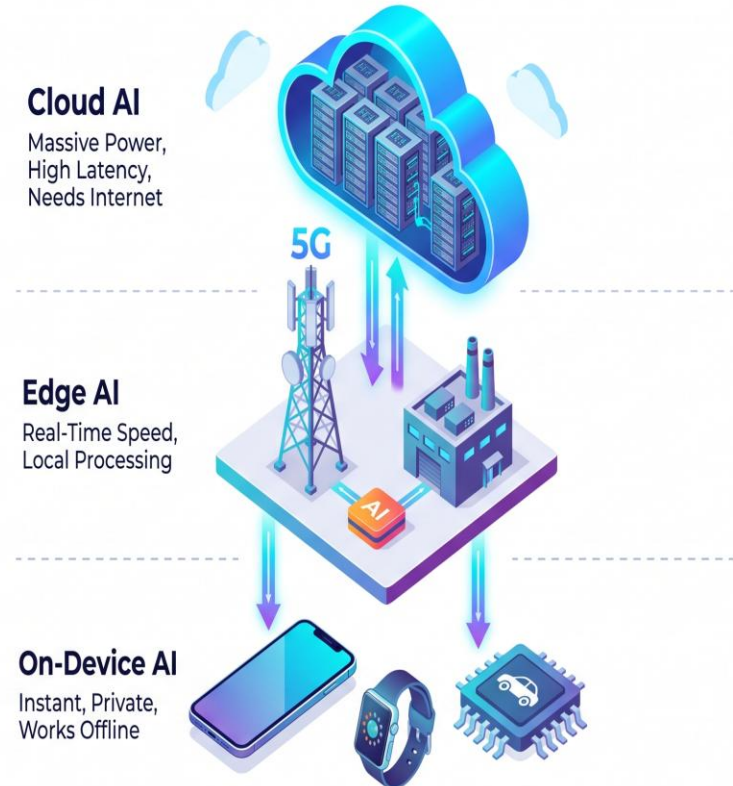
📶 Edge AI (Local Hub)

- What:** Small servers or 5G towers close to the user.
- Pros:** Real-time speed, saves bandwidth.
- Cons:** Less powerful than the cloud.

📱 On-Device AI (The Pocket Brain)

- What:** AI running directly on your phone or watch.
- Pros:** Instant, totally private, works offline.
- Cons:** Very limited computing power.

AI on Devices, Edge, and Cloud



06

Python Orientation

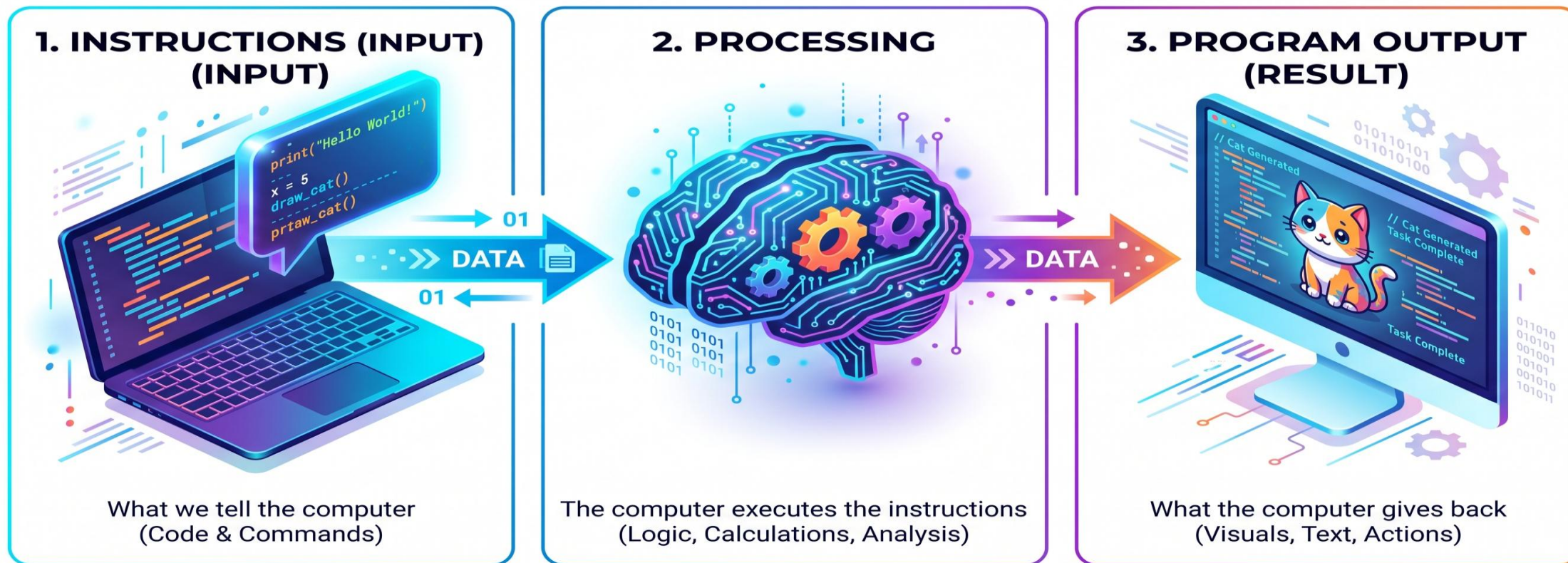
Huawei HCCDA-AI

- Easy to Learn:** Simple, English-like syntax. Developers focus on AI logic, not complex coding rules.
- Rich Libraries:** Pre-built "toolboxes" (PyTorch, TensorFlow, Scikit-learn, Pandas). No need to code from scratch.
- Massive Community:** Millions of developers. Tons of free tutorials, forums, and ready-made code.
- Great for Data:** Best tools for cleaning, analyzing, and visualizing data (Pandas, NumPy).
- Flexible & Fast:** Acts as "glue." Easily connects with C/C++ for heavy math, making it both easy to write and fast to run.

Why Python is Used in AI



HOW PROGRAMMING WORKS: INPUT, PROCESSING, OUTPUT



SOURCE / FURTHER READING

Thank You

Any Question?